HOMEWORK 2, WEEK 2

$g(p,Q)=\int_0^1 p(x)Q(x)dx \quad \mathbb{R}_2[x]\times\mathbb{R}_2[x]\to\mathbb{R}$

① g is inner product if:

- symmetric $g(p,Q)=\int_0^1 p(x)Q(x)\,dx=\int_0^1 Q(x)p(x)\,dx=g(Q,p)$ ✓
- linear $g(\alpha_1 p_1+\alpha_2 p_2,Q)=\int_0^1(\alpha_1 p_1+\alpha_2 p_2)(x)\,Q(x)\,dx=$

$=\int_0^1(\alpha_1 p_1(x)Q(x)+\alpha_2 p_2(x)Q(x))\,dx=\alpha_1\int_0^1 p_1(x)Q(x)\,dx+\alpha_2\int_0^1 p_2(x)Q(x)\,dx=$

$=\alpha_1 g(p_1,Q)+\alpha_2 g(p_2,Q)$ left linear $\wedge$ symmetric $\to$ right linear ✓

- positive definite

non-negative $g(p,p)=\int_0^1 p(x)^2\,dx\geq 0$

definite $\int_0^1 p(x)^2dx=0\to p(x)=0$ because polynomials are continuous functions ✓

② orthonormal basis β for $\mathbb{R}_2[x]$ wrt. g:

$g(b_i,b_j)=\delta_{ij}$

start from the standard basis $\{1,x,x^2\}$

$U_1=1$

$Q_1=\frac{1}{\sqrt{\int_0^1 1^2dx}}=\frac{1}{\sqrt{1}}=\underline{\underline{1}}$

$U_2=x-\int_0^1 x\cdot 1\,dx\cdot 1=x-\frac{1}{2}$

$\|U_2\|=\sqrt{\int_0^1(x-\frac{1}{2})^2dx}=\sqrt{\frac{1}{12}}$

$Q_2=\underline{\underline{\sqrt{3}(2x-1)}}$

$U_3=x^2-\int_0^1 x^2\cdot 1\,dx\cdot 1-\int_0^1 x^2\cdot\sqrt{3}(2x-1)\,dx\cdot\sqrt{3}(2x-1)=\dots=x^2-x+\frac{1}{6}$

$\|U_3\|=\sqrt{\int_0^1(x^2-x+\frac{1}{6})dx}=\sqrt{\frac{1}{180}}$

$Q_3=\underline{\underline{\sqrt{5}(6x^2-6x+1)}}$

③ $\alpha(p) = \int_0^2 p(x)\,dx \quad \mathbb{R}_2[x] \to \mathbb{R}$

· linear $\alpha(ap+bq) = \int_0^2 (ap+bq)(x)\,dx = \int_0^2 (ap(x)+bq(x))\,dx =$

$= \int_0^2 ap(x)\,dx + \int_0^2 bq(x)\,dx = a\int_0^2 p(x)\,dx + b\int_0^2 q(x)\,dx = \underline{a\alpha(p)+b\alpha(q)}$

④ r such that $\alpha(p) = g(p,r)$

Riesz theorem (with the Hilbet matrix):

$$H = \begin{bmatrix} g(1,1) & g(1,x) & g(1,x^2) \\ g(x,1) & g(x,x) & g(x,x^2) \\ g(x^2,1) & g(x^2,x) & g(x^2,x^2) \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$$

$$\alpha = \begin{bmatrix} \alpha(1) \\ \alpha(x) \\ \alpha(x^2) \end{bmatrix} = \begin{bmatrix} \int_0^2 1\,dx \\ \int_0^2 x\,dx \\ \int_0^2 x^2\,dx \end{bmatrix} = \cancel{\begin{bmatrix} 1 \\ \frac{1}{2} \\ \frac{1}{4} \end{bmatrix}} \begin{bmatrix} 2 \\ 2 \\ \frac{8}{3} \end{bmatrix}$$

unknown $r = \begin{bmatrix} c \\ b \\ a \end{bmatrix}$

$$Hr = \alpha$$

$$r = H^{-1}\alpha = \begin{bmatrix} 9 & -36 & 30 \\ -36 & 192 & -180 \\ 30 & -180 & 180 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ \frac{8}{3} \end{bmatrix} = \begin{bmatrix} 26 \\ -168 \\ 180 \end{bmatrix}$$

$\sim r(x) = 180x^2 - 168x + 26$

STATEMENT: This is my work. More detailed calculations for part 2 are scribbled on another paper. I used Gemini to verify my results and to explain Riesz theorem.